Technische Hochschule Ingolstadt

(Forecasting)

Time series: Step Count(Part 1) and Microgrid Energy (Part 2)

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Notice on the Use of AI Assistance

In this seminar paper, the AI-supported language model ChatGPT by OpenAI as well as Gemini by Google were used. The models served to support the formulation and structuring of the manuscript. All suggestions generated by ChatGPT were carefully reviewed and adapted so that they reflect the author's own way of thinking as well as the desired style of expression and refer exclusively to verifiable sources.

The final responsibility for the content and the scientific integrity of the submitted work lies entirely with the author. All content generated by ChatGPT and Gemini was thoroughly reviewed and independently revised to meet academic requirements. The present work is the intellectual property of the author and fulfills the principles of academic independence and authenticity. All formulations and decisions can be justified upon request. Apart from the explicitly mentioned tools, no further aids or sources were used.

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Part 1: Self-Collected Data Analysis

1 Introduction of Self-Collected Dataset

1.1 Description of What Was Measured

The self-collected dataset contains daily step count data, represented by the variable "Steps per Day", recorded over a two-month period. The measurements were obtained using Apple's Health app on an iPhone 15 Pro Max, which automatically and continuously tracks the number of steps taken each day. The unit of measurement is steps. The dataset has a temporal resolution of one observation per day. Overall, the dataset contains 61 daily observations covering the period from March 1, 2026 to April 30, 2026.

1.2 Rationale for Dataset Choice

Daily step counts were selected because they represent a practical and easily understandable personal time series. The data is collected continuously in everyday life and can therefore be used to analyze real behavioral patterns over time.

This dataset is suitable for forecasting because step counts may contain temporal structures such as weekly patterns or differences between weekdays and weekends. For example, physical activity can vary depending on daily routines, university schedules, workdays, free time, or planned exercise. Therefore, the dataset provides a meaningful basis for investigating trends, variation, and possible recurring patterns.

The dataset is also relevant from a real-world perspective. Step counts are commonly used in health and fitness applications to monitor physical activity and support personal goal tracking. Forecasting future activity levels can help identify changes in mobility behavior and support a better understanding of personal habits.

Overall, the dataset was chosen because it is simple to collect, directly related to daily behavior, and suitable for applying basic time series analysis and forecasting methods.

2 Data Collection Method

2.1 Description of the Data Collection Procedure

The data was collected automatically with the iOS Health app. The app records daily step counts in the background while the iPhone is carried during everyday activities. The observation period covered 61 days, from March 1, 2026 to April 30, 2026. Since the app summarizes the number of steps per calendar day, the dataset contains one observation per day.

The data collection was passive and did not require manual input during the observation period. The values reflect regular daily movement under normal everyday conditions. No additional activities were planned only for the purpose of changing the dataset.

After the observation period, the data was exported from Apple Health as an XML file and then converted into CSV format for analysis in Python (Apple Inc., 2026). A limitation of this method is that the measurement depends on carrying the iPhone consistently. If the phone is not carried, some steps may not be recorded. This limitation was considered when interpreting the results.

2.2 Tools, Devices, or Software Used

The data was collected with the Apple Health app on an iPhone 15 Pro Max running iOS 26.5. The app uses the phone's built-in motion sensors, especially the accelerometer and gyroscope, to estimate the number of steps taken during the day. The step count

is aggregated by the app over a 24-hour period and resets at midnight according to the phone's default time settings. No smartwatch, fitness tracker, or additional external device was used. This ensures that all recorded values come from the same device and the same measurement process.

For the analysis, the exported data was converted into CSV format and processed in Python. This made it possible to clean the dataset, check missing values, create visualizations, and analyze temporal patterns.

3 Descriptive Analysis

3.1 Basic Overview: Summary, Visualization, Missing Values, Outliers

To begin the descriptive analysis, the daily step count data from March 1, 2026 to April 30, 2026 was examined using summary statistics, visual inspection, missing-value checks, and outlier detection. The dataset contains 61 daily observations with one observation per calendar day. The central variable is the number of steps per day.

Statistic	**Value (Steps)**
Mean	12,686
Median	12,299
Standard deviation	1,892
Minimum	10,143
Maximum	17,251

Table 1: Descriptive statistics of the daily step count dataset (March 1–April 30, 2026).

The descriptive statistics show that the daily step counts are generally high and relatively consistent. The mean value is 12,686 steps and the median is 12,299 steps, which indicates that the distribution is not strongly distorted by extreme observations. The standard deviation of 1,892 steps shows that there is still noticeable day-to-day variation, but the variability is moderate compared to the overall activity level. The lowest observed value is 10,143 steps, while the highest value is 17,251 steps.

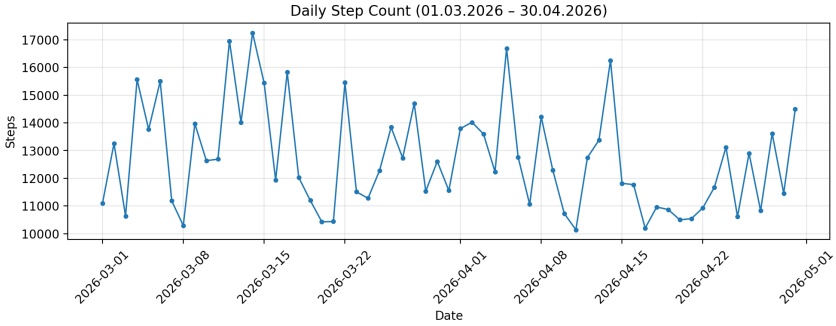


Figure 1: Daily step count from March to April 2026. The time series shows regular day-to-day variation without a strong visual break or extreme drop.

The time series plot in Figure 1 shows that the observations fluctuate from day to day, but most values remain within a relatively narrow range between about 10,000 and 15,000 steps. Several higher days are visible, especially in mid-March, but these values remain plausible for normal daily activity and do not indicate obvious measurement errors. The dataset was also checked for missing values and missing timestamps. No missing values were found in either the date column or the step count column. In addition, the complete calendar range from March 1 to April 30, 2026 is present, meaning that no daily timestamps are missing. Therefore, no imputation or correction of missing data was necessary. Outliers were examined using the interquartile range (IQR) method. The first quartile is 11,092 steps and the third quartile is 13,840 steps, resulting in an IQR of 2,748 steps. Based on the standard 1.5-IQR rule, the lower outlier boundary is 6,970 steps and the upper boundary is 17,962 steps. Since all observed values lie within this interval, no statistical outliers were detected.

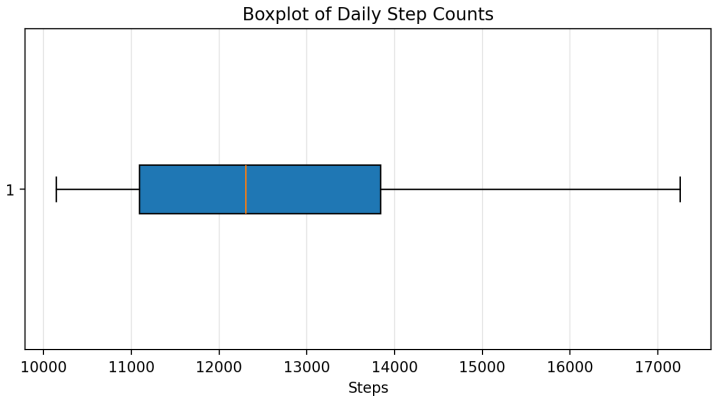


Figure 2: Boxplot of daily step counts. The distribution shows moderate spread and no statistical outliers according to the IQR criterion.

The boxplot in Figure 2 supports this result. The distribution has a moderate spread and no individual point is visible beyond the whiskers. This suggests that even the highest values are still realistic within the observed personal activity pattern. Overall, the basic data quality is strong: the series is complete, chronologically consistent, and does not require outlier removal.

3.2 Serial Dependence and Temporal Structure Analysis

To investigate whether the daily step counts depend on previous days, autocorrelation (ACF) and partial autocorrelation (PACF) plots were created. These plots help assess whether past observations contain predictive information for later observations.

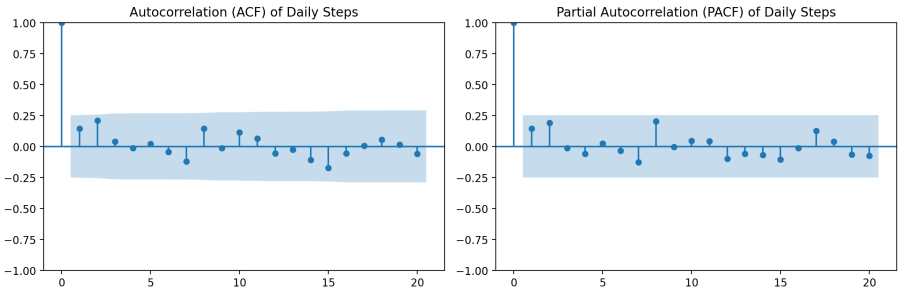


Figure 3: Autocorrelation and partial autocorrelation plots of daily step counts. Most lags remain within the confidence bounds, indicating weak serial dependence.

The ACF plot shows only weak positive autocorrelation at the first lags. Lag 1 has an autocorrelation of approximately 0.14, and lag 2 is approximately 0.21. These values are small and remain within the confidence interval. The PACF plot leads to a similar interpretation: there is no clearly dominant lag that would indicate a strong autoregressive structure. This suggests that the number of steps on one day only weakly explains the number of steps on the following day. Such behavior is plausible for personal activity data, because daily movement is affected by routines, appointments, weather, motivation, and other external factors.

To examine weekly patterns, the data were grouped by weekday. The average step count is highest on Thursday, with approximately 13,335 steps. Sunday follows with about 13,003 steps, while Saturday has the lowest average with approximately 12,192 steps. The differences between weekdays are visible but not extreme.

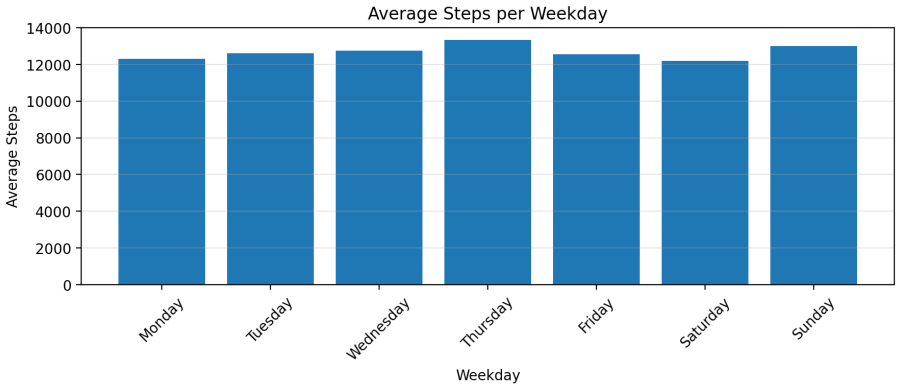


Figure 4: Average number of steps per weekday. Thursday has the highest average step count, while Saturday has the lowest average.

The weekday pattern in Figure 4 indicates a mild weekly structure. However, the differ-

ence between weekdays and weekends is small overall. Weekdays have an average of 12,711 steps, while weekends have an average of 12,621 steps. This means that the dataset does not show a strong weekend drop. Instead, activity remains relatively high throughout the week.

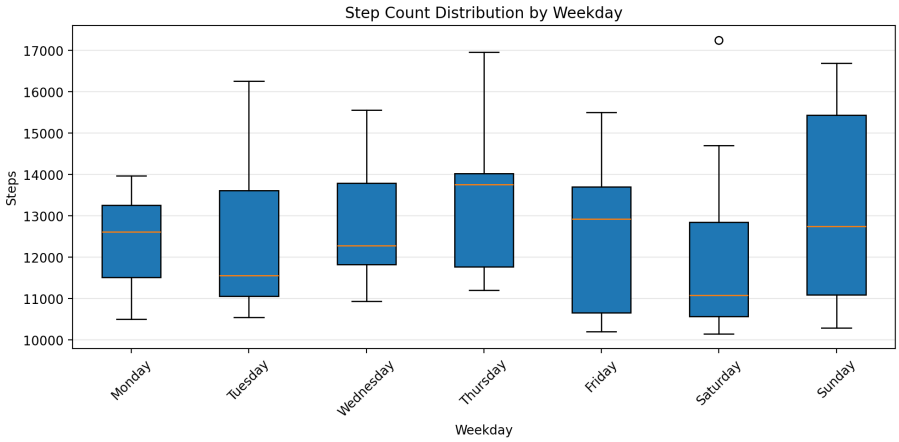


Figure 5: Step count distribution by weekday. The boxplots show that variability differs between weekdays, with wider spread on Saturday and Sunday.

The weekday boxplots in Figure 5 provide additional information about variability. Saturday and Sunday show a wider spread than some weekdays, which suggests that weekend activity is less predictable. This is realistic because weekend behavior is often more flexible and may include either rest days or longer walks.

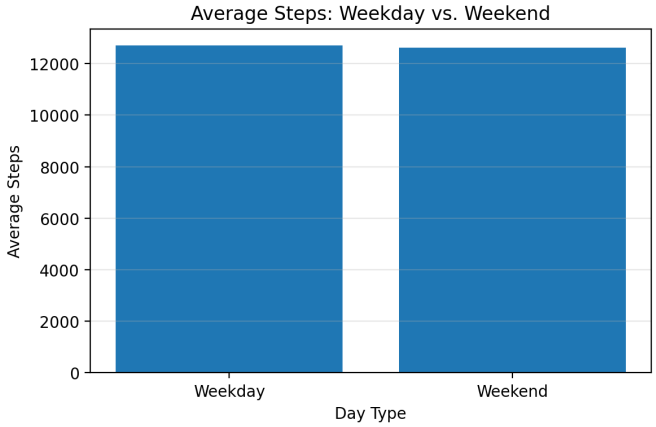


Figure 6: Average step count on weekdays versus weekends. The difference is small, suggesting no strong weekend effect.

A monthly comparison was also conducted. March has an average of approximately 13,020 steps, whereas April has an average of approximately 12,342 steps. This indicates a slight decrease from March to April. However, the observation period covers only two months, so this difference should not be interpreted as a reliable seasonal pattern.

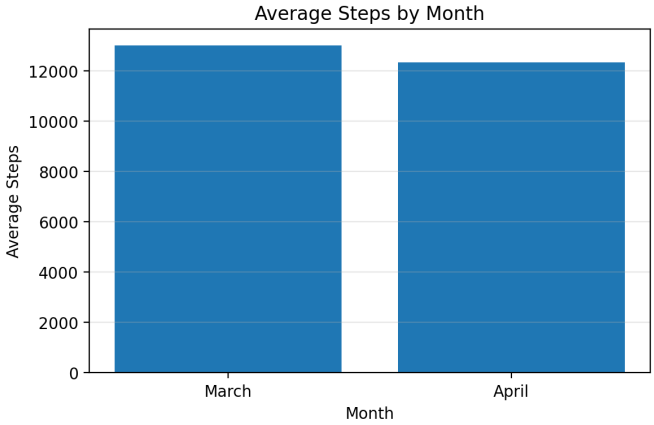


Figure 7: Average step count by month. March shows a slightly higher average than April, but the two-month window is too short for robust monthly seasonality conclusions.

A lag plot was created to further assess the relationship between consecutive observations. The scatterplot compares each daily step count with the value from the previous day.

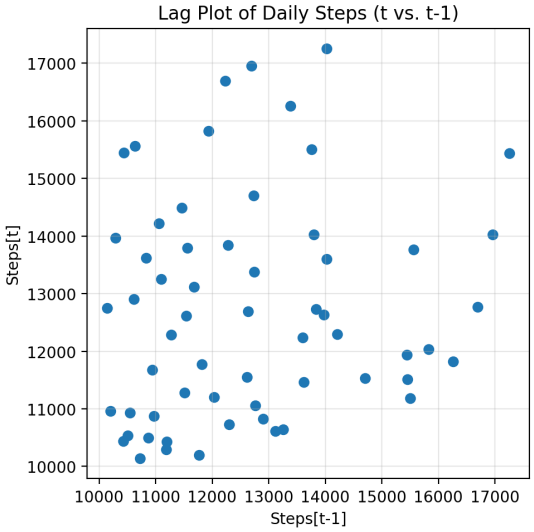


Figure 8: Lag plot of daily step counts. The diffuse scatter pattern indicates weak linear dependence between consecutive days.

The lag plot in Figure 8 shows a broad scatter of points rather than a clear diagonal pattern. This confirms the ACF and PACF results: consecutive days are not strongly linearly related. Therefore, a simple autoregressive interpretation would not be sufficient to describe the full behavior of the series. Finally, a 7-day rolling mean and rolling standard deviation were plotted to examine short-term changes in level and variability.

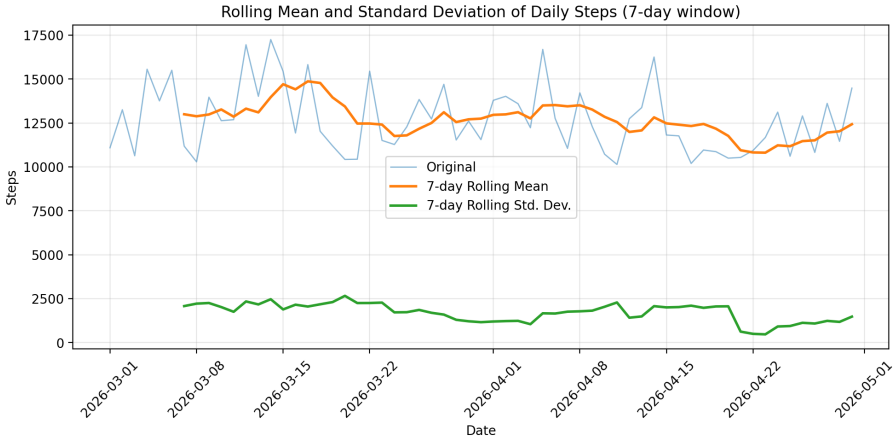


Figure 9: Rolling mean and rolling standard deviation of daily step counts using a 7-day window. The rolling mean fluctuates moderately, while the rolling standard deviation changes over time.

The rolling mean remains within a relatively stable range but shows moderate fluctuations across the observation period. The rolling standard deviation also changes over time, indicating that some weeks are more variable than others. Nevertheless, the changes are not extreme, and the overall visual impression is more stable than strongly trending.

3.3 Pattern Identification, Decomposition, and Stationarity Check

To separate trend, seasonal, and residual components, STL decomposition was applied to the daily step count series with a weekly seasonal period of seven days. This choice is appropriate because the data are daily and potential behavioral cycles are expected to repeat weekly.

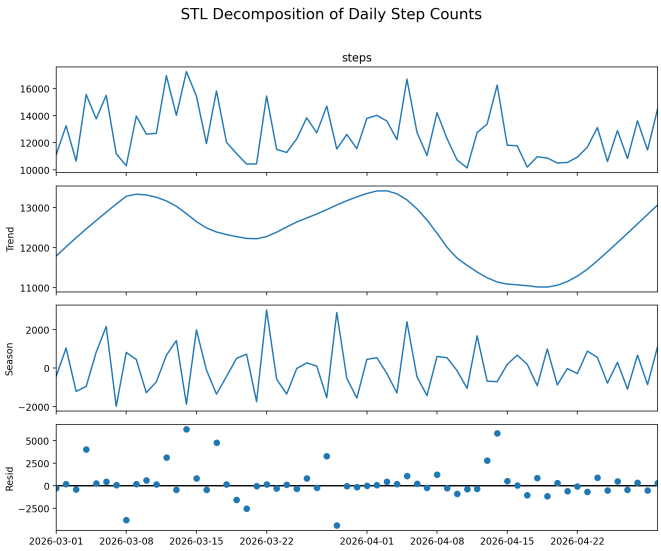


Figure 10: STL decomposition of daily step counts. The decomposition separates the observed series into trend, weekly seasonal component, and residual component.

The STL trend component shows only a weak trend structure. Numerically, the approximate trend strength is 0.133, which indicates that the trend explains only a limited part of the total variation. Although the trend component rises toward the end of the period, the overall two-month window is too short to claim a stable long-term upward trend. In addition, the monthly averages show that March is slightly higher than April, which argues against interpreting the data as a clear continuous increase.

The seasonal component reflects a possible weekly rhythm, but it is not dominant. The calculated seasonal strength is 0.000 using the applied variance-based measure, meaning that weekly seasonality is weak compared to the residual day-to-day variation. This does not mean that weekday effects are absent, but it does mean that the weekday pattern is not strong enough to explain a large part of the overall variability. Stationarity was assessed visually using the rolling mean and rolling standard deviation and formally using the Augmented Dickey-Fuller (ADF) test. The rolling plots show some variation in mean and standard deviation over time, but the fluctuations do not suggest a strong persistent trend. The ADF test produced a test statistic of -3.999 and a p-value of 0.0014. Since the p-value is below 0.05, the null hypothesis of a unit root can be rejected. Therefore, the daily step count series can be considered stationary in the statistical sense. From a forecasting perspective, this has important implications. Since strong non-stationarity is not detected, differencing may not be necessary as a first preprocessing step. However, the weak weekly structure and the behavioral nature of the data suggest that models should still consider weekday information or seasonal components if forecasting is later performed.

4 Discussion

Interpretation of Basic Data Properties

The self-collected step count data shows a generally high and stable level of daily activity. The mean value of 12,686 steps and the median of 12,299 steps are close to each other, indicating that the distribution is not strongly skewed or dominated by extreme values. The standard deviation of 1,892 steps shows moderate day-to-day variation. In practical terms, this means that daily activity usually fluctuates by around 1,900 steps around the average level, which is realistic for personal movement behavior.

The dataset is complete, as no missing values or missing dates were detected. In addition, the IQR-based outlier analysis did not identify any statistical outliers. This suggests that all recorded observations are plausible and can be retained for the analysis. Overall, the data quality is strong and no correction or imputation was necessary.

Interpretation of Temporal Structures

The ACF and PACF plots indicate only weak serial dependence. Most lags remain within the confidence bounds, and no dominant autoregressive pattern is visible. This means that the step count of one day does not strongly predict the step count of the following day.

The weekday analysis shows a mild weekly structure. Thursday has the highest average step count, while Saturday has the lowest. A possible explanation is that Thursday may reflect a more active weekday routine, such as commuting, university activities, or planned exercise. In contrast, Saturday may be more variable because weekend routines are less fixed and can include either rest days or longer outdoor activities.

The difference between weekdays and weekends is small, with 12,711 steps on weekdays and 12,621 steps on weekends. Therefore, the data does not show a clear weekend drop. The wider spread on Saturday and Sunday suggests that weekend activity is more flexible and less predictable. The monthly comparison shows that March has a slightly higher average step count than April. Since the dataset covers only two months, this difference should be interpreted cautiously and not as a reliable seasonal pattern.

Interpretation of Global Patterns: Trend, Seasonality, and Stationarity

The STL decomposition shows only a weak trend component. The calculated trend strength is 0.133, meaning that the trend explains only a small part of the total variation. Although the trend rises slightly toward the end of the period, the short observation window does not allow a reliable long-term trend interpretation.

The weekly seasonal component is also weak. The calculated seasonal strength is 0.000, which suggests that weekday effects exist but do not explain much of the overall variation. Day-to-day behavioral factors appear to be more important than a fixed weekly rhythm.

The ADF test supports stationarity, with a test statistic of -3.999 and a p-value of 0.0014. Since the p-value is below 0.05, the null hypothesis of a unit root can be rejected. Therefore, the series can be considered stationary in the statistical sense.

For future forecasting, this means that differencing is probably not required as a first preprocessing step. The weak autocorrelation and stationarity suggest that a simple ARIMA-type model could be tested, but the lack of strong temporal patterns may limit its accuracy. Models that include weekday effects or external regressors, such as weather, schedule information, or planned exercise, would likely improve forecast performance.

Part 2: Public Dataset Analysis

5 Dataset Overview and Exploratory Data Analysis

5.1 Introduction and Topic Relevance

The public dataset used in Part 2 is the *Microgrid Energy Forecasting Dataset*, obtained from Kaggle (Colabsss, 2024). The forecasting target of this analysis is load demand, which represents the electricity demand within the microgrid. This variable is relevant because demand forecasting is a central task in energy management and supports short-

term operational planning.

The dataset contains time series data related to the operation of a microgrid system, including variables that describe energy demand, generation, storage, and grid interaction. The original observations are recorded at a 15-minute frequency and therefore provide a detailed view of short-term energy behavior.

Microgrids are particularly interesting for forecasting because their operation often depends on fluctuating energy sources, changing consumption patterns, and possible interaction with battery storage or the external grid. Reliable load forecasts can reduce uncertainty and help balance supply and demand more efficiently. If future demand can be estimated more accurately, the system can react more efficiently to expected load changes.

In the notebook, the raw 15-minute data was first validated and then converted to hourly frequency. This resampling step was chosen to reduce very short-term noise and create a more manageable time series while still preserving the main intraday demand structure. All subsequent analyses and models in Part 2 are therefore based on hourly load demand values. The goal is to examine the temporal structure of the demand series and compare the forecasting performance of Holt-Winters, SARIMA, and Prophet.

5.2 Interpretation of Basic Data Properties

After loading the public microgrid dataset, the timestamp column was converted into a datetime format and used as the time index. The original dataset contains 5,787 observations and was recorded at a 15-minute frequency. It covers the period from January 1, 2023 at 00:00 to March 2, 2023 at 06:30. Since the forecasting analysis is conducted on an hourly basis, the original 15-minute observations were aggregated to hourly values. After this resampling step, the time series contains 1,447 hourly observations from January 1, 2023 at 00:00 to March 2, 2023 at 06:00.

The forecasting target is load demand. To understand the basic structure of this target variable, descriptive statistics were calculated for the hourly load demand series.

Descriptive statistic	**Hourly load demand**
Number of observations	1,447
Mean	446.49
Median	448.24
Standard deviation	102.74
Minimum	139.72
Maximum	738.29

Table 2: Descriptive statistics of the hourly load demand series.

The descriptive statistics show that the mean and median are very close to each other. The mean hourly load demand is approximately 446.49, while the median is approximately 448.24. This indicates that the distribution is relatively balanced around its central level.

The standard deviation of 102.74 shows that the series contains noticeable variation over time, which is expected for electricity demand data.

The minimum value of 139.72 and the maximum value of 738.29 indicate substantial short-term fluctuations. Such variation is plausible in a microgrid context, since electricity demand can change depending on time of day, consumption behavior, and operational conditions. Therefore, the spread of the values should not automatically be interpreted as poor data quality.

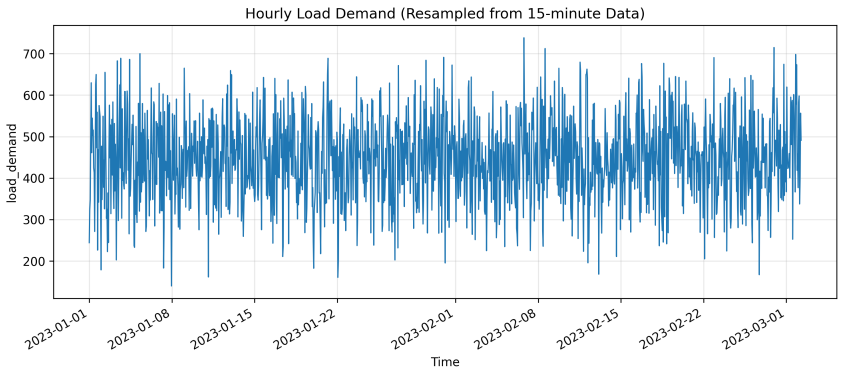


Figure 11: Hourly load demand over the full observation period after resampling from 15-minute data.

Figure 11 shows the hourly load demand over the full observation period. The series fluctuates strongly, but no obvious structural break or data gap is visible. This confirms that the main challenge of the dataset is not missing data, but the high volatility of the target variable.

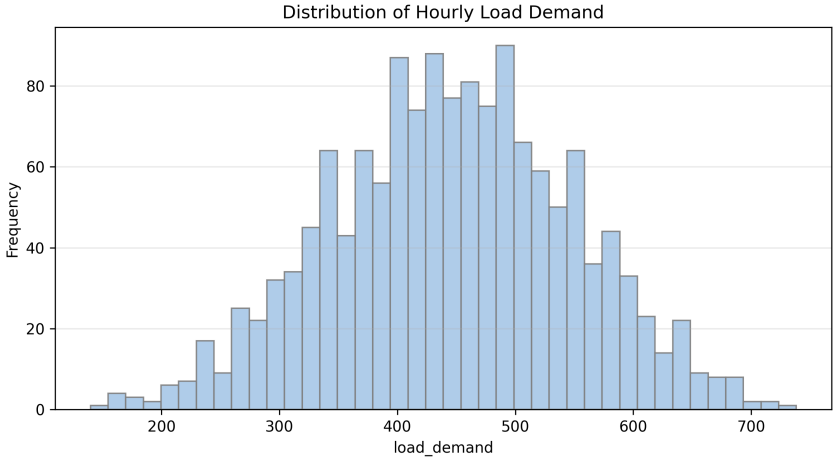


Figure 12: Distribution of hourly load demand values.

The distribution in Figure 12 confirms that most hourly demand values are concentrated around the central range of the series. However, some lower and higher values are also present, reflecting the variability of the energy demand pattern.

Missing values and missing timestamps were also checked. No missing hourly timestamps were detected after resampling, and the target series did not contain missing observations. Therefore, no imputation or artificial filling of missing values was necessary before modeling.

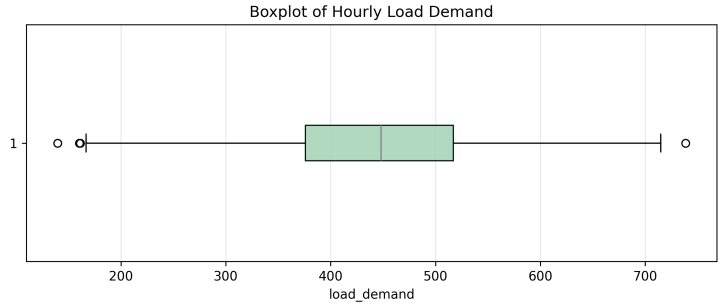


Figure 13: Boxplot of hourly load demand values.

The boxplot in Figure 13 provides an additional view of the distribution. Using the interquartile range method, four potential outliers were identified in the hourly load demand series. These observations were not removed because short-term peaks or drops can represent realistic demand situations in an energy system. Keeping these values preserves the original behavior of the microgrid demand series.

Overall, the basic data properties indicate that the dataset is suitable for time series forecasting. It has a regular hourly structure, no missing timestamps, a clearly defined target variable, and realistic demand variability. The main analytical challenge is therefore not data cleaning, but modeling the short-term volatility of load demand.

5.3 Analysis of Temporal Structures

The temporal structure of the hourly load demand series was examined by analyzing patterns by hour of day, weekday, and daily average demand. Since the dataset was resampled to hourly frequency, the hour of day is especially relevant for detecting potential intraday demand behavior.

First, the average hourly load demand was calculated for each hour of the day. This allows an assessment of whether certain hours systematically show higher or lower demand.

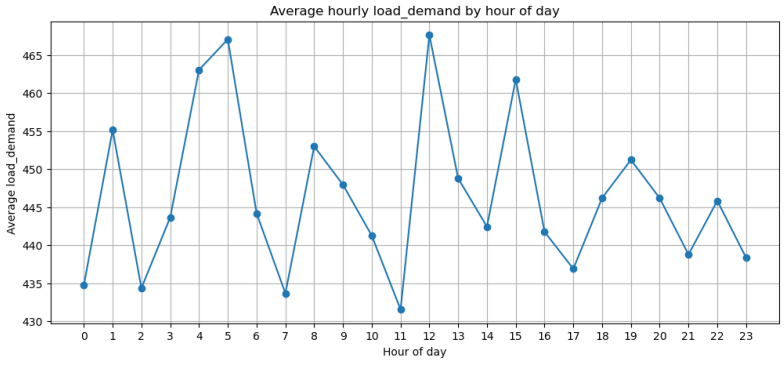


Figure 14: Average hourly load demand by hour of day.

Figure 14 shows that the average load demand differs across the hours of the day, but the pattern is not completely smooth. The highest average hourly demand occurs around hour 12, with approximately 467.64, while the lowest average value occurs around hour 11, with approximately 431.55. The difference between these two values indicates that intraday variation exists, but it is moderate compared to the overall variability of the series. A possible explanation is that the microgrid may serve a mixed-use environment with a relatively stable baseload, where electricity demand remains present throughout the day. The minor midday peak could be related to higher daytime activity, heating or cooling demand, or operational processes during working hours.

In addition to the hourly pattern, weekday effects were examined. The average hourly load demand is highest on Wednesday, with approximately 450.96, and lowest on Saturday, with approximately 435.84. The difference between weekdays is therefore visible but not very large. This suggests that the dataset does not show a strong weekday-weekend contrast. From a microgrid perspective, this may indicate that the connected demand sources operate relatively consistently across the week. The slightly lower Saturday value could reflect reduced activity in some facilities or lower operational demand during parts of the weekend.

Daily average demand was also analyzed to investigate whether the general demand level changes over the observation period. For this step, only complete days with 24 hourly observations were used. This avoids distortions from the final incomplete day in the dataset, which ends at 06:00.

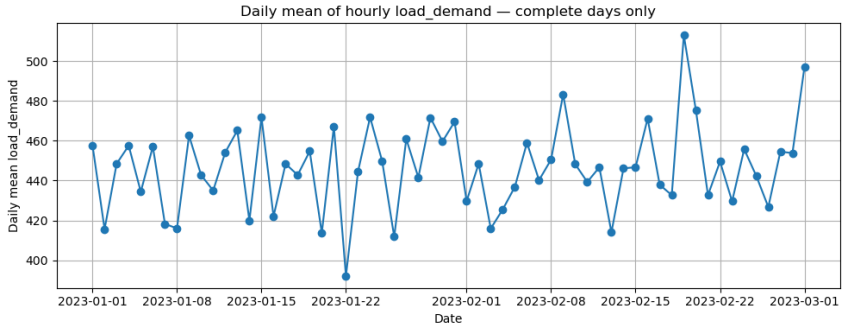


Figure 15: Daily mean of hourly load demand using complete days only.

Figure 15 shows that the daily mean load demand fluctuates over time, but without a clearly persistent upward or downward trend. Across the complete daily observations, the average daily mean is approximately 446.34, with a standard deviation of approximately 21.83. The lowest daily mean occurs on January 22, 2023, with approximately 392.13, while the highest daily mean occurs on February 19, 2023, with approximately 512.93.

Overall, the temporal analysis indicates that the load demand series contains short-term variation and moderate temporal patterns. The hour-of-day analysis points to some intraday structure, while the weekday analysis suggests only limited differences between days of the week. The daily mean series fluctuates over the observation period, but does not provide evidence of a strong global trend. These findings support the use of forecasting models that can handle short-term variability and potential daily seasonality.

5.4 Analysis of Global Patterns (Trend, Seasonality, Stationarity)

To examine the global structure of the hourly load demand series, trend, seasonality, autocorrelation, and stationarity were analyzed. These aspects are important because they help determine whether the series contains systematic long-term movements, recurring patterns, or statistical instability over time.

First, an STL decomposition was applied to the hourly load demand series (Cleveland et al., 1990). Since the data was analyzed at hourly frequency, a seasonal period of 24 was used to represent potential daily seasonality.

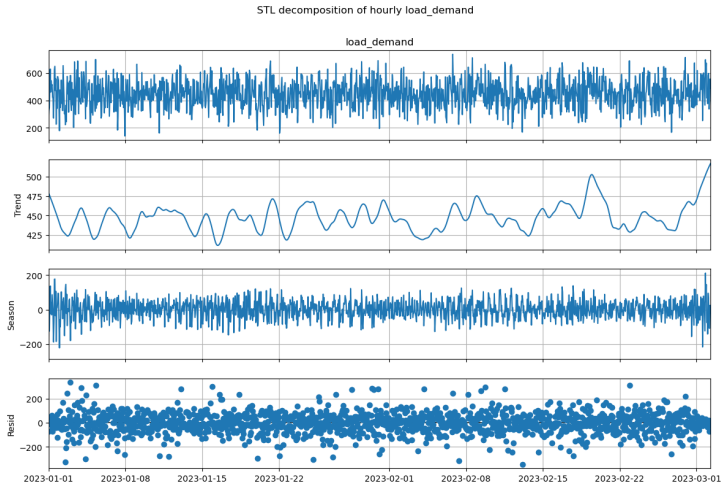


Figure 16: STL decomposition of the hourly load demand series using a daily seasonal period of 24 hours.

Figure 16 separates the observed load demand series into trend, seasonal, and residual components. The trend component fluctuates around the general level of the series, but it does not show a clear persistent upward or downward movement over the full observation period. This indicates that the dataset does not contain a strong global trend. The calculated trend strength is approximately 0.049, which confirms that the trend explains only a small part of the total variation.

The seasonal component shows recurring intraday variation, but it is not dominant. The calculated seasonal strength is approximately 0.284. This suggests that daily seasonality is present to some extent, but a large part of the variation remains in the residual component. In other words, the demand series contains some daily structure, while still being strongly influenced by short-term fluctuations that are not fully explained by the seasonal component.

To further assess the temporal dependence of the series, autocorrelation and partial autocorrelation plots were created. The ACF shows the correlation between the series and its past values, while the PACF shows the direct relationship with previous lags after controlling for shorter lag effects.

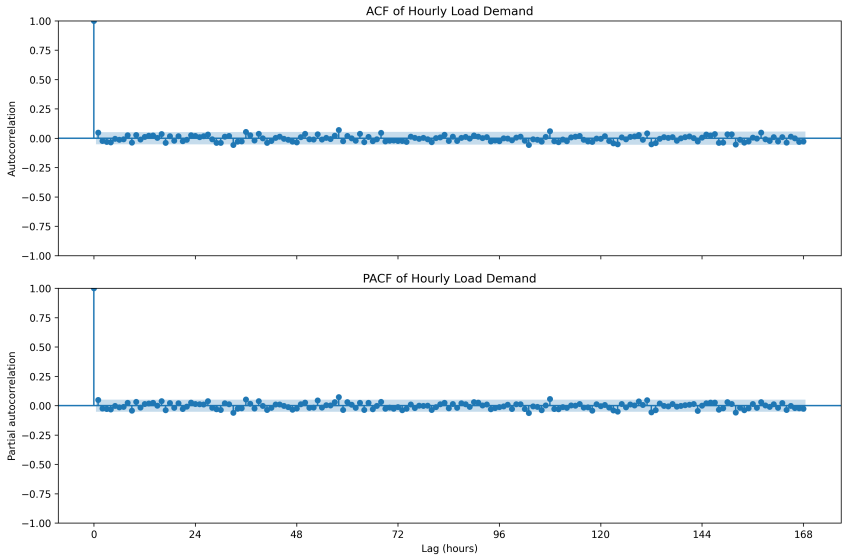


Figure 17: ACF and PACF plots of the hourly load demand series.

Figure 17 indicates that autocorrelation in the hourly load demand series is generally weak. The first lag has only a small autocorrelation of approximately 0.05, and the autocorrelation at the daily seasonal lag of 24 hours is also low, at approximately 0.02. The PACF leads to a similar interpretation, as no clearly dominant lag structure is visible. This suggests that past hourly values alone explain only a limited part of future load demand.

From a modeling perspective, the weak ACF and PACF patterns do not suggest a strong high-order autoregressive or moving-average structure. Therefore, a relatively low-order SARIMA specification is appropriate. The seasonal period of 24 was still retained because the exploratory analysis showed moderate daily seasonality, even though the autocorrelation at lag 24 itself is weak.

To further assess the stability of the series over time, a rolling mean and rolling standard deviation were calculated using a 24-hour window. This window corresponds to one full day in the hourly dataset.

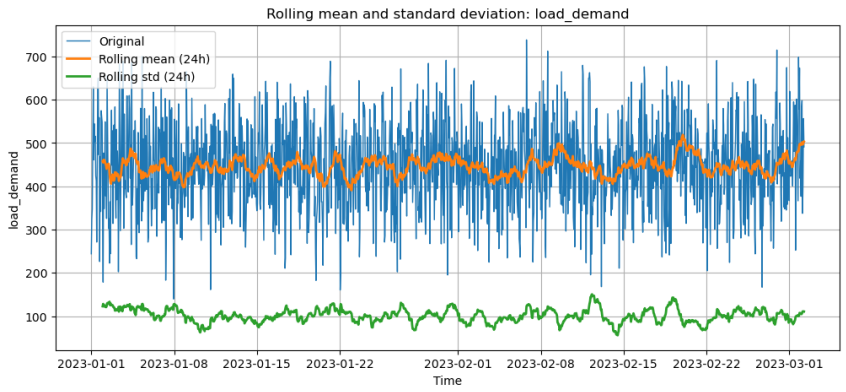


Figure 18: Rolling mean and rolling standard deviation of hourly load demand using a 24-hour window.

Figure 18 shows that the rolling mean remains within a relatively stable range over time, although short-term fluctuations are visible. The rolling standard deviation also changes over the observation period, but it does not indicate a clear structural break. This supports the interpretation that the series is volatile, but not strongly non-stationary in its overall level.

Stationarity was also examined formally using the Augmented Dickey-Fuller test (Dickey and Fuller, 1979). The original hourly series produced a p-value below 0.001. Since this value is below the 5% significance level, the null hypothesis of a unit root can be rejected. Therefore, the hourly load demand series can be considered stationary in the statistical sense.

In the notebook, the ADF test was also applied to the first-differenced series and the seasonal-differenced series with a lag of 24 hours. Both versions also produced p-values below 0.001. This confirms that the stationarity conclusion is robust, although differencing is not strictly required for the original hourly series.

Overall, the global pattern analysis shows that the series has only a weak trend component, moderate daily seasonality, and weak autocorrelation. The stationarity test suggests that differencing is not strictly required before modeling the original hourly series. However, the presence of some daily seasonal structure supports the use of forecasting models that can account for seasonal patterns, such as Holt-Winters, SARIMA, and Prophet.

6 Methodology

6.1 Model Selection

The forecasting task in Part 2 focuses on the hourly load demand series of the microgrid dataset. The objective is to compare classical and interpretable time series forecasting approaches under the same data conditions. Therefore, three models were selected: Holt-Winters exponential smoothing, SARIMA, and Prophet.

The three models represent different forecasting perspectives. Holt-Winters is a smoothing-based model that estimates level, trend, and seasonal components using exponential smoothing weights. SARIMA is a statistical time series model that combines autoregressive, differencing, moving-average, and seasonal structures. Prophet is an additive forecasting model that represents a time series as a combination of trend, seasonality, and optional event effects (Hyndman and Athanasopoulos, 2021; Taylor and Letham, 2018).

This model selection is connected to the exploratory analysis. The hourly load demand series showed only a weak global trend, moderate daily seasonality, and high short-term variability. Therefore, it was useful to compare a seasonal smoothing model, a seasonal ARIMA-based model, and an additive decomposition-based model. All three models were applied to the same target variable, load demand, and evaluated on the same chronological test set.

No exogenous variables were included in the forecasting models. As a result, the com-

parison focuses on how well each model can forecast future load demand using only the historical behavior of the target series. This is important because the broader dataset contains additional variables, but the present analysis intentionally evaluates univariate forecasting performance.

6.2 Justification of Chosen Models

The Holt-Winters model was selected because the hourly load demand series showed moderate daily seasonality. In an additive Holt-Winters model, the observed value is represented through a level component, a trend component, and a seasonal component. The one-step-ahead forecast can be written in simplified form as:

$$\hat{y}_{t+1} = l_t + b_t + s_{t+1-m} \tag{1}$$

In this equation, l_t represents the level, b_t represents the trend, s_t represents the seasonal component, and m represents the seasonal period. For this dataset, the seasonal period was set to $m = 24$, because the data was analyzed at hourly frequency and 24 observations represent one full day. In the notebook, Holt-Winters was implemented with additive trend and additive seasonality. This is appropriate because the seasonal variation was interpreted as an absolute change around the load demand level rather than as a multiplicative percentage effect (Hyndman and Athanasopoulos, 2021).

SARIMA was selected because it is designed for time series that may contain autocorrelation, differencing requirements, and seasonal patterns. A general SARIMA model can be written as:

$$\Phi_P(B^m)\phi_p(B)(1-B)^d(1-B^m)^D y_t = \Theta_Q(B^m)\theta_q(B)\varepsilon_t \tag{2}$$

In this equation, the parameters p, d, and q describe the non-seasonal autoregressive, differencing, and moving-average orders. The parameters P, D, and Q describe the corresponding seasonal orders. The term m represents the seasonal period. In this analysis, the implemented model used the non-seasonal order $(1, 1, 1)$ and the seasonal order $(1, 0, 1, 24)$. The seasonal period of 24 reflects the daily cycle in the hourly dataset. The ACF and PACF plots did not suggest a strong high-order lag structure, which supports the use of a relatively low-order SARIMA specification. At the same time, the moderate daily seasonality justified retaining a seasonal period of 24 (Hyndman and Athanasopoulos, 2021).

Although the implementation uses the `SARIMAX` class from the `statsmodels` library, no exogenous regressors were included. Therefore, the model corresponds methodologically to a univariate SARIMA model. This distinction is important because the dataset contains additional variables related to generation, storage, and grid interaction, but these variables were not used as external predictors in this analysis.

Prophet was selected as a third model because it provides a different modeling perspective. Prophet represents a time series as an additive model:

$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t \tag{3}$$

In this formulation, the term $g(t)$ represents the trend, $s(t)$ represents seasonal effects, and $h(t)$ represents holiday or event effects. The final error term captures random variation that is not explained by the trend or seasonal components (Taylor and Letham, 2018). In the present analysis, Prophet was used with daily and weekly seasonality, while yearly seasonality was disabled because the dataset covers only about two months. No holiday component was included, so the model mainly focused on trend and seasonal effects.

Overall, the three selected models allow a structured comparison between smoothing-based, statistical, and additive forecasting approaches. This is suitable for the microgrid load demand series because the data contains moderate daily seasonality, weak autocorrelation, and strong short-term fluctuations. The goal is therefore not only to identify the lowest error metric, but also to compare how different forecasting model families handle the same hourly demand series.

6.3 Preparation for Forecasting

Before modeling, the dataset was prepared in several steps. First, the timestamp column was converted into a datetime format and used as the index of the dataset. The original observations were recorded at a 15-minute frequency. Since the analysis and forecasting task are conducted on an hourly basis, the dataset was resampled to hourly frequency. Numeric variables were aggregated using the mean, while categorical variables were handled using the most frequent available value within each hourly interval.

After resampling, the hourly dataset contained 1,447 observations. The modeling series was then defined as the hourly load demand target. The time index was checked for consistency, and no missing target observations remained in the final hourly series. This step was important because the selected forecasting models require a continuous chronological structure.

The dataset was split chronologically into training and test data. This is necessary for time series forecasting because future observations must not be used during model training. The first 80% of the observations were used for training, while the final 20% were used as the test set. This resulted in 1,157 training observations and 290 test observations. The training period covers January 1, 2023 at 00:00 to February 18, 2023 at 04:00. The test period covers February 18, 2023 at 05:00 to March 2, 2023 at 06:00.

Several modeling assumptions were considered before fitting the forecasting models. The hourly resampling assumes that the mean load demand within each hour is an appropriate representation of the original 15-minute observations. The chronological train-test split assumes that future observations are unknown during training and therefore avoids data leakage. For Holt-Winters, the main assumption is that the series can be represented through additive level, trend, and seasonal components with a daily seasonal period of 24 hours. For SARIMA, the stationarity results and the weak ACF/PACF structure support the use of a low-order seasonal specification, although the model still assumes that the selected lag structure captures the main temporal dependence. For Prophet, the main assumption is that the time series can be decomposed into additive trend and seasonal components. Since no exogenous variables were included, all models assume that future

load demand can be estimated only from the historical behavior of the target series.

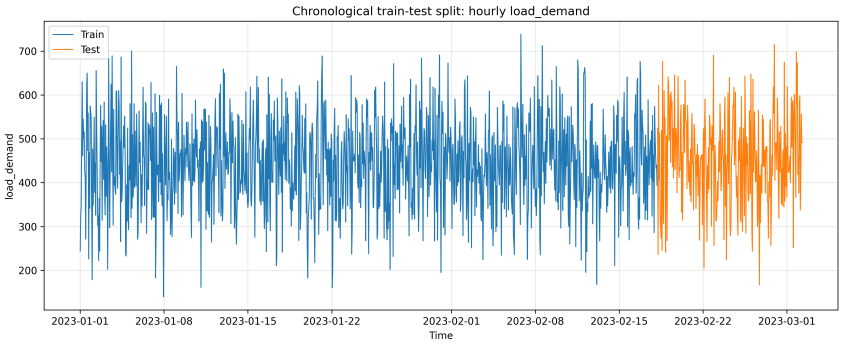


Figure 19: Chronological train-test split of the hourly load demand series.

Figure 19 illustrates the chronological separation between training and test observations. The models were fitted only on the training period and then used to forecast the test period. This ensures that the evaluation reflects an out-of-sample forecasting situation.

The forecast accuracy was assessed using three error metrics: mean absolute error, mean squared error, and root mean squared error. MAE measures the average absolute forecast error and is easy to interpret in the unit of the target variable. MSE penalizes larger errors more strongly because the errors are squared. RMSE is the square root of MSE and is again expressed in the same unit as the original target variable. Using all three metrics provides a more complete view of model performance, especially because high load demand fluctuations can lead to larger forecast errors.

The same evaluation procedure was applied to all models. After each model produced forecasts for the test period, residual diagnostics were also conducted. The residuals were inspected over time, their distribution was visualized, autocorrelation was examined, and the Ljung-Box test was used to check whether residual autocorrelation remained. This made it possible to evaluate not only the numerical forecast accuracy, but also the remaining error structure of each model.

7 Results

7.1 Model Implementation

The three forecasting models were implemented on the same hourly load demand series and trained on the same chronological training set. The test set was not used during model fitting and was only used for evaluating out-of-sample forecast performance.

The Holt-Winters model was implemented using exponential smoothing with an additive trend and additive seasonality. Since the dataset was analyzed at hourly frequency, the seasonal period was set to 24 in order to represent one daily cycle. The model parameters were optimized automatically during fitting.

The SARIMA model was implemented using the `SARIMAX` class from the `statsmodels` library. However, no exogenous regressors were included. Therefore, the model corresponds

methodologically to a univariate SARIMA model. The implemented specification used the non-seasonal order $(1, 1, 1)$ and the seasonal order $(1, 0, 1, 24)$. The seasonal period of 24 again represents the daily structure of the hourly time series.

Prophet was implemented as an additive forecasting model with daily and weekly seasonality enabled. Yearly seasonality was disabled because the dataset covers only about two months, which is not sufficient to estimate yearly seasonal patterns reliably. The Prophet model was trained using the required `ds` and `y` format, where `ds` represents the timestamp and `y` represents the hourly load demand.

All models produced forecasts for the same 290 observations in the test period. This allows a direct comparison of the models under identical data conditions.

7.2 Residual Diagnostics

After generating the forecasts, residual diagnostics were conducted for each model. The residuals were calculated as the difference between the actual test values and the predicted values. This step is important because forecast accuracy metrics alone do not show whether systematic error patterns remain in the model residuals.

The residual summaries show that all three models have residual means close to zero. Holt-Winters has a mean residual of approximately -6.98, SARIMA has a mean residual of approximately 11.87, and Prophet has a mean residual of approximately 7.25. This indicates that none of the models shows a strong average bias over the full test period. The residual standard deviations are also very similar across the three models, ranging from approximately 104.51 to 105.75.

The Ljung-Box test was used to examine whether residual autocorrelation remained at the daily lag of 24 hours. For all three models, the p-values are clearly above the 5% significance level. Holt-Winters has a p-value of approximately 0.795, SARIMA has a p-value of approximately 0.894, and Prophet has a p-value of approximately 0.866. Therefore, the null hypothesis of no residual autocorrelation cannot be rejected. This suggests that the residuals do not contain strong remaining autocorrelation at the daily lag.

Overall, the residual diagnostics indicate that the three models behave similarly. The residuals are centered approximately around zero and do not show strong remaining autocorrelation. However, the residual standard deviations remain relatively high. This confirms that the hourly load demand series contains substantial short-term variability that is difficult to forecast precisely with univariate models.

7.3 Forecast Results

The forecast performance was evaluated using mean absolute error, mean squared error, and root mean squared error. These metrics were calculated on the same test period for all models.

Model	MAE	MSE	RMSE
Holt-Winters	84.34	11193.45	105.80
SARIMA	84.31	11024.63	105.00
Prophet	83.88	11027.36	105.01

Table 3: Forecast accuracy comparison on the hourly load demand test set.

The results show that the three models perform very similarly. Prophet achieves the lowest MAE with a value of approximately 83.88. This means that, on average, Prophet has the smallest absolute deviation from the actual test values. However, SARIMA achieves the lowest MSE and RMSE, with an RMSE of approximately 105.00. Prophet follows very closely with an RMSE of approximately 105.01, while Holt-Winters has a slightly higher RMSE of approximately 105.80.

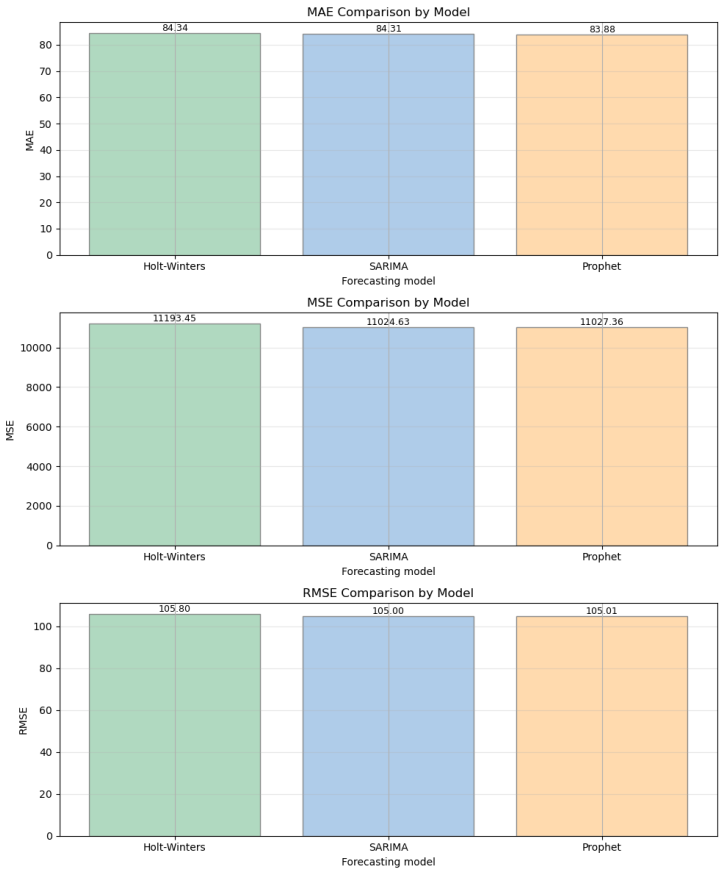


Figure 20: Comparison of MAE, MSE, and RMSE for Holt-Winters, SARIMA, and Prophet.

Figure 20 confirms that the differences between the models are small. SARIMA performs marginally best according to RMSE, while Prophet performs marginally best according to MAE. Since the numerical differences are very limited, no model clearly dominates the others. The results should therefore be interpreted cautiously rather than as strong evidence for one superior model.

7.4 Visualizations

The forecast comparison plot provides a visual interpretation of the model performance on the test period. It compares the actual hourly load demand values with the forecasts produced by Holt-Winters, SARIMA, and Prophet.

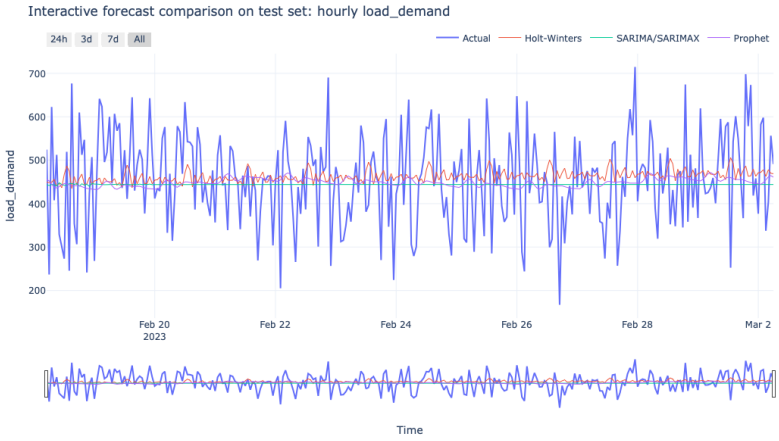


Figure 21: Forecast comparison of actual load demand and model predictions on the test set.

Figure 21 shows that all three models follow the general level of the test series, but they do not fully capture the strong short-term fluctuations. The actual load demand values contain frequent peaks and drops, while the model forecasts are smoother. This behavior is expected because the models are based only on the historical target series and do not include external explanatory variables.

The visual comparison also supports the numerical results. The forecasts of SARIMA and Prophet are very close to each other, and Holt-Winters follows a similar overall pattern. The main forecast errors occur during rapid changes in demand. This indicates that the main difficulty of the forecasting task is not estimating the average level of demand, but predicting short-term peaks and troughs.

Overall, the results show that all three models provide reasonable forecasts for the general demand level, but none of them fully captures the high volatility of the hourly load demand series. The small differences in MAE, MSE, and RMSE suggest that the choice between the three models should not be based only on the numerical ranking. Model interpretability, implementation simplicity, and the potential inclusion of external variables should also be considered.

8 Discussion

The results show that hourly load demand in the microgrid dataset is difficult to forecast precisely with univariate time series models. Holt-Winters, SARIMA, and Prophet were all able to follow the general level of the series, but none of the models fully captured the strong short-term fluctuations in the test period.

The three models performed very similarly. Prophet achieved the lowest MAE, while SARIMA achieved the lowest MSE and RMSE. However, the differences were very small, so no model can be considered clearly superior. This suggests that the main challenge is not estimating the average demand level, but predicting random short-term peaks and drops.

The residual diagnostics support this interpretation. The residual means were close to zero, and the Ljung-Box test did not indicate strong remaining autocorrelation at the daily lag. However, the residual standard deviations remained high, showing that substantial short-term variability was not fully explained by the models.

A key limitation is that no exogenous variables were included. The models only used past load demand values, although the dataset also contains variables related to generation, storage, and grid interaction. Including such variables in a SARIMAX model or another multivariate forecasting approach could improve future forecasts. In addition, selecting the SARIMA order was challenging because the ACF and PACF plots did not show a clear dominant lag structure.

From a practical microgrid perspective, the forecasts could support short-term planning of battery usage, grid import, and generation scheduling. However, the RMSE of approximately 105 shows that forecast uncertainty remains relevant. Therefore, operators would still need reserve capacity for demand spikes that the models cannot predict accurately.

Overall, the selected models are useful for estimating the general demand level, but less effective at predicting rapid short-term changes. Future work could improve performance by using longer historical data and additional explanatory variables.

9 Personal Reflection

Working on this project helped me gain a better understanding of time series analysis and forecasting. By using both a self-collected step count dataset and a public microgrid energy dataset, I learned how different temporal datasets can be prepared, analyzed, and interpreted.

The first part showed me how important careful data collection and preprocessing are. Even a simple personal dataset such as daily step counts requires checks for missing values, outliers, temporal patterns, and possible limitations. I also realized that external factors, such as routines, weather, or appointments, can influence the data but are not always available for modeling.

In the second part, I gained practical experience with forecasting models such as Holt-Winters, SARIMA, and Prophet. I learned that model performance should not be judged by one metric alone, since different metrics can lead to slightly different rankings. The results also showed that similar error values should be interpreted carefully instead of assuming that one model is clearly superior.

Overall, this project strengthened my understanding of the complete forecasting workflow,

from data preparation and exploratory analysis to model evaluation and interpretation. It also showed me the limits of univariate forecasting and the potential value of including additional explanatory variables in future analyses.

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